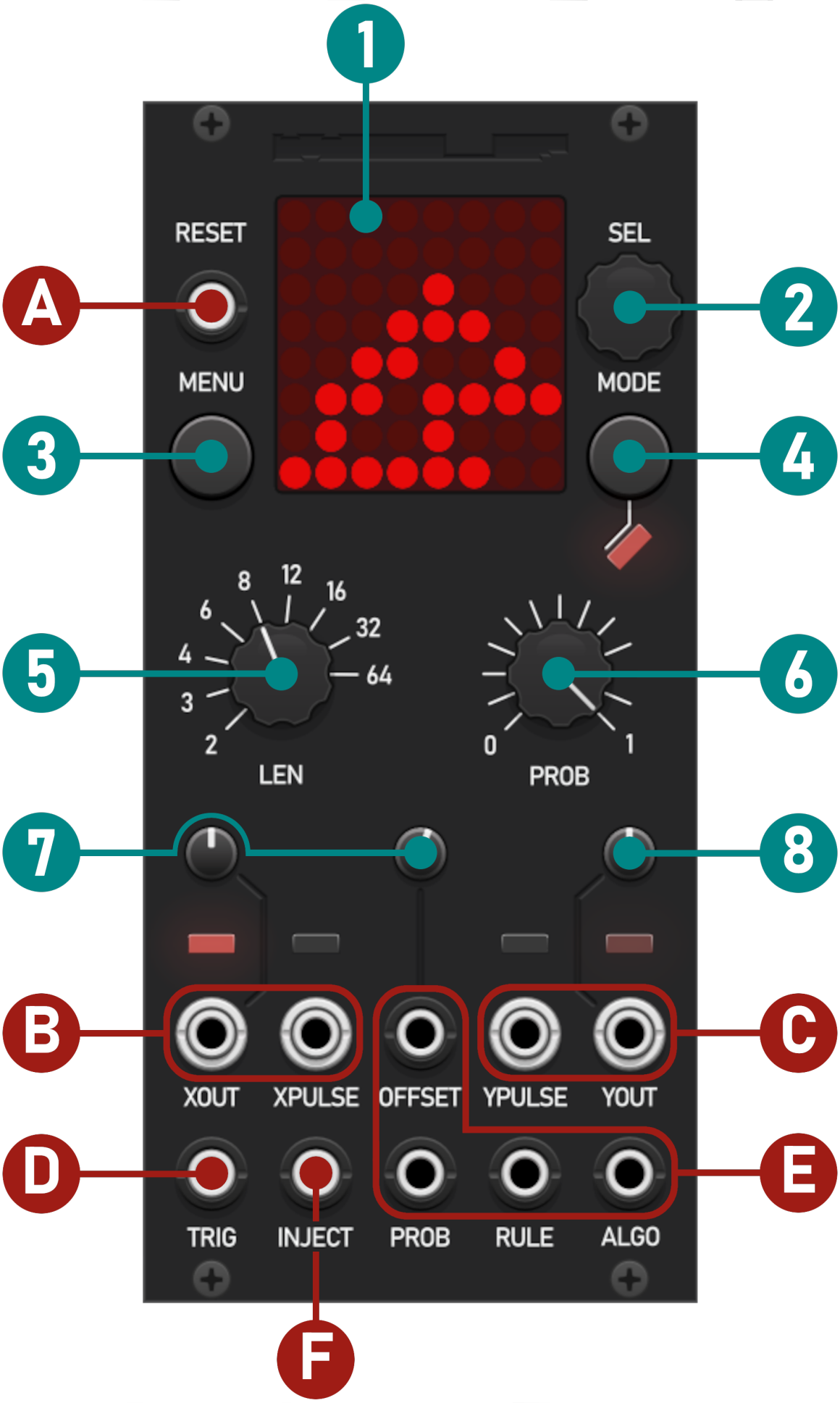
WOLFRAM

USER MANUAL


1
RESET
SEL
A
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MENU
MODE
3
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12
16
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LEN
PROB
7
8
B
C
XOUT
XPULSE
OFFSET
YPULSE
YOUT
D
E
TRIG
INJECT
PROB
RULE
ALGO
F

About Wolfram

Wolfram is a multi-algorithmic sequencer capable of generating complex patterns emerging from simple sets of rules. When triggered at audio rate, the module can act as a digital oscillator with a timbre determined by the algorithm. At any moment, a sequence can be captured and locked into a loop of a given length—a concept developed by the iconic Turing Machine from Music Thing Modular.

Controls

1. 8x8 cell **Matrix** and **Menu** display.
2. Multi-purpose **Select** encoder. Rotate to change the **Rule**. When **Menu** is active, controls the displayed parameter (**Seed**, **Mode**, **Slew**, or **Algorithm**). Double-click to trigger **Reset**; when adjusting a parameter, double-click restores it to its default value.
3. Toggle the **Menu**.
4. Algorithm-dependent **Mode** select, see the **Algorithms** section for details on each mode. When the **Menu** is open, cycles through the menus pages.
5. Set the sequence **Length**.
6. Controls the **Probability** that a new generation will occur or the sequence will loop. At the **0** position, the sequence is locked and loops at the set **Length**. At the **1** position, the active **Algorithm** generates new steps freely.
7. Applies a horizontal **Offset** to the **Matrix**, affecting the output without altering the internal sequence.
8. Sets the **X** and **Y** output voltage range independently (0-10Vpp). Does not affect pulse outputs.

Inputs and Outputs

A. If the sequence is locked, triggering **Reset** returns the step position to 0. If the module is generating, **Reset** pushes the **Seed** into the **Algorithm**.
B. C. **X** and **Y** algorithm-dependent voltage and pulse outputs. See the **Algorithms** section for details on how these values are determined. If **Audio Rate** mode is active, the output voltage is passed through a DC-removing filter.
D. Triggers the module to advance one step if locked, or generate a new step. If **Audio Rate** mode is active, triggers are derived from zero-crossings.
E. Control voltage inputs for **Offset**, **Probability**, **Rule**, and **Algorithm**.
F. Send a positive trigger to **Inject** a living cell into the **Matrix**, or a negative trigger to remove a cell.

Context Menu

The context menu provides options to toggle **Audio Rate** mode, quick select an **Algorithm**, adjust **Display** and **Cell** styles, and toggle **Sync**. When **Sync** is active, control changes are applied at the next step; otherwise, they take effect immediately.

Wolf Algorithm

The Wolf algorithm is a classic 8-bit one-dimensional automata. A row of cells is fed into one of 256 rules, determining the next iteration. While this is a simple mechanism, it can lead to a variety of complex, repetitive, fractal, and random sequences.

Some iconic rules include Rule 30, known for its chaotic structures reminiscent of natural phenomena . Rule 90, featuring a Sierpiński triangle; and Rule 110, which is capable of universal computation.

Mode

CLIP. Dead cells border the row.
WRAP. Loops the row's boundaries to the opposite side.
RAND. Random cells border the row.

Output

X OUT. The bottom row converted to an 8-bit value.
Y OUT. The right column converted to an 8-bit value.
X PULSE. High if bottom left cell is living.
Y PULSE. High if top right cell is living.

Life Algorithm

The Life algorithm exists on a two-dimensional 8×8 plane. Choose a rule that defines this space, creating a pocket dimension where bacterial colonies multiply and mazes grow. Among these rules, Conway's Game of Life is perhaps the best-known, renowned for its ability to produce complex patterns, from pulsars and oscillators to spaceships. However, this is just one of thirty selected rules, each unique in character.

Much of the Life algorithm's behaviour is inherently unpredictable. To make it easier to navigate to useful results, a range of seeds are available. These include different types of randomness, as well as specific starting patterns that have been found to produce interesting evolutions when paired with compatible rules. Try different combinations to see how each rule responds.

Mode

CLIP. Dead cells surround the plane.
WRAP. Loops the plane's boundaries to the opposite side.
BOTL. Wraps the plane like a Klein Bottle.
RAND. Random cells surround the plane.

Output

X OUT. Population of living cells.
Y OUT. The 8x8 plane as a 64-bit value.
X PULSE. High if population increases.
Y PULSE. High if population dies or becomes static, can be routed to **Reset**.